A-A-53653
19 August 1988

COMMERCIAL ITEM DESCRIPTION

CEMENT POWDER, ZINC OXIDE, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a dental zinc oxide cement powder.

Salient characteristics:

General. The powder shall be mixable with and compatible with eugenol, USP. The powder when mixed with eugenol, in accordance with manufacturer's instructions shall produce a cement. The cement shall be of the non-staining type and shall be suitable for use in luting and hermetically sealing silver or gutta percha points in endodontically treated teeth. The cement shall be hygroscopic, radiopaque, water insoluble, and shall not discolor tooth structure.

Material. The composition of the dental cement powder shall be in accordance with Table I. One hundred percent of the powder shall pass through a 100 mesh (150 micrometer) sieve in accordance with ASTME 11 and tested in accordance with USP.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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TABLE I. Powder composition.

Ingredient	Specification	Percentage
Zinc Oxide	United States Pharmacopeia (USP)*	38.6 ± 2.0
Partially hydrogenated rosin	FDA approved for food packing and processing operations.	28.8 ± 2.5
Bismuth Subnitrate	USP	14.4 ± 1.4
*Barium Sulfate	USP	14.4 ± 1.4
Sodium Borate	NF	3.8 ± 1.0

*Except for arsenic, Maximum limit for arsenic shall be 0.0002%.

Performance. The powder when mixed with eugenol according to manufacturer's instructions shall produce a cement. The cement shall exhibit the following properties.

Working time and appearance. The cement shall be a smooth, creamy, homogeneous paste of light cream color and shall have a working time of 15 to 20 minutes when tested as specified in test section under working time and appearance.

Hardening time. The cement shall have a tough, rigid body and shall be fractured rather than pierced when tested as specified in test section under hardening time.

Adhesiveness and solubility. The cement shall firmly adhere to a glass mixing slab and dissolve in chloroform when tested as specified in test section under adhesiveness and solubility.

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Irritation. The cement shall be non-toxic and non-irritating. Unless otherwise specified, prior to delivery the manufacturer shall furnish a statement directly to the Directorate of Medical Materiel, Technical Services Branch, DPSC-RST, Defense Personnel Support Center, 2800 S. 20th Street, Philadelphia, Pa 19101-8419 stating that the material does not normally irritate periapical tissues, does not contain poisonous ingredients in sufficient concentration to be harmful to human beings and provides a hermetic seal in the pulp canal when used as directed.

Manufacturer's instructions. Adequate and accurate instructions for mixing and use of the material shall accompany each unit package. These instructions shall include the following as a minimum requirement:

Powder/liquid ratio in terms of number of drops of liquid to grams of powder.

Method of mixing, including armamentaria necessary for use.

Spatulating time required and its effect on setting time.

Workmanship. The powder shall be free from lumps and foreign matter and any defect which detracts from its appearance or impair its serviceability.

Unit. Jar (JR). One jar, containing 100 grams of powder with manufacturer's instructions, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one jar of cement powder. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests shall be conducted at inspection level S-1 and acceptance number shall be zero.

Examination. The dental cement powder shall be examined to determine compliance with all requirements of this specification. Examination shall be in accordance with, but shall not be limited to the defects listed below.

Defects

Powder not free of lumps or foreign matter.

Powder not free of contamination.

Cap not on tight.

Cap cannot be removed by hand.

Instructions missing.

Tests. Tests shall be conducted to determine compliance with specification requirements.

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Working time and appearance. Spatulate the powder with eugenol, USP, for 4 minutes on a clean, dry, glass mixing slab in accordance with manufacturer's instructions and under the following conditions: Temperature 22 to 24° Celsius; Relative humidity 40 to 45 percent. Four and one-half minutes after starting to spatulate, peak the mixture with a suitable instrument which is wiped clean before each insertion. Continue to peak at 1/2-minute intervals and observe for slumping. Working time shall be calculated as the span of time from the start of spatulation to the 1/2-minute interval preceding the formation of the first permanent peak. Peak shall be considered permanent when it does not slump to its original level within 1/2-minute.

Hardening time. The test samples, prepared in accordance with the working time and appearance test specified herein, shall be allowed to stand for 10 hours at constant temperature and humidity and shall be tested for hardness with a sharp instrument.

Adhesiveness and solubility. The test samples, prepared in accordance with the hardening test specified herein, shall adhere so firmly to the glass slab that they can be chipped off only with a knife or other sharp instrument. The test samples shall then be readily removable from the slab by the use of chloroform and a wiping cloth.

Contractor certification. The contractor shall certify that the product offered meets the the salient characteristics of this description and conform to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE 1: The following National Stock Number is covered by this document:

Stock Number	Item Identification
6520-01-062-7694	CEMENT POWDER, ZINC OXIDE, DENTAL 100 Gram

Note 2: The following National Stock Number represents eugenol used to mix with the powder to form the cement:

Stock Number	Item Identification
6505-00-153-8379	Eugenol, USP, 1 oz (28.35 Gram)

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Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD - MB

Agent:

DLA - DM

Civil Agency Coordinating Activities:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2124

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53653	CEMENT POWDER, ZINC OXIDE, DENTAL

3 NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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